# INDEX

## PART-A


| Ex No | CONTENTS | Page No. | REMARKS |
|---|---|---|---|
| 01 | Create C# console application which reads a set of strings and then prints the string having maximum number of vowels. | | |
| 02 | Write C# console program which stores different types of topics with the values Topics = {"Introduction to C#", "Variables", "Data Types", "Loops", "If statements", "jump statements", "classes and objects", "Inheritance", "Constructors"} Using switch case statement categorise the topics as "Basic", "Control Flow" and "OOPs Concepts". | | |
| 03 | Write a console application to generate a number pattern in the shape of a right angled triangle with the numbers increased by 1. | | |
| 04 | Create a C# console application that receives the following information from a set of students: Student Id, Student Name, Course Name, and Date of Birth. The application should also display the information of all the students once the data is entered. Implement this using an array of objects. | | |
| 05 | Create a windows form application that allows the user to enter a number in the textbox named ‘getnum’. Check whether the number in the textbox ‘getnum’ is palindrome or not. Print the message accordingly in the label control named lbldisplay when the user clicks on the button ‘check’. | | |
| 06 | Implement C# windows form application to perform money conversion. | | |
| 07 | Design a webpage of a hotel which displays different Menu as per the Time of Visit. | | |

| PART-B | | | |
|---|---|---|---|
| **Ex No** | **CONTENTS** | **Page No.** | **REMARKS** |
| 01 | Create an MDI form with several child forms. (at least 3 child forms with proper information) MDI parent form should contain menu strip as shown below with menu options to open, close and rearrange the child forms.<br>**Child Forms** **Window**<br>Open Cascade<br>Close Tile Horizontal<br>Tile vertical<br>Arrange icons. | | |
| 02 | Design and implement a windows form application in C# to calculate and display an employee's salary details using the given conditions.<br>If Basic <= 20000 D.A is 40% of Basic, H.R.A is 10% of Basic, P.F 12% of Gross,<br>PT is Rs.100<br>If Basic> 20000 D.A is 50% of Basic, H.R.A 15% of Basic, P.F 12% of Gross,<br>PT is Rs.150<br>Gross = Basic+D.A +HRA, Net = Gross -PT –PF<br>The form should allow the user to input Employee ID, Employee Name, Date of Birth and Basic Pay using appropriate controls (Textbox/DateTimePicker).<br>On clicking "Calculate" button, the application should display the calculated values in appropriate output controls (ReadOnlyTextBoxes or Labels). | | |
| 03 | Design windows form application to order pizza in a restaurant. Select pizza size using radiobuttons, toppings using check boxes and quantity using numericUpDown control. PictureBox should display an image of the pizza with selected toppings.<br>Buttons :<br>i)) Order – To calculate and display the total amount and order summary.<br>ii)Another pizza variety – Clears earlier size, toppings and quantity selections in the form so that<br>you can add another variety of pizza to the current order.<br>iii)Clear : cancels the order. | | |
| 04 | Create a web form for Login Module which adds a username and password to the database. The username in the database should be a primary key. Also implement proper authentication functionality. | | |
| 05 | Create a web application to design an admission form with client side validations. | | |
| 06 | Design a webpage to enter Student information such as Student no, Student Name, marks in 3 subjects. Use the following buttons for, | | |

| | | | |
|---|---|---|---|
| | • Add -> for adding the record to the database (Insert at least 5 records). Calculate total, percentage, grade and store it.<br>• Display – Display the records from the database using data grid view. | | |
| 07 | Create ASP.NET web application with the given user interface to input two strings(str1,str2) and perform two operations “Search” and Construct” by clicking respective buttons. The result of both operations shall be displayed in output TextBox.<br><br>SERACH: If the user clicks on “Search” button then appearance of str2 is searched in str1 and removed from str1. Also, the characters of str1, before and after str2 are concatenated together. Ignore the cases where there is no character in str1 before or after the str2.<br><br>CONSTRUCT: If user clicks on “Construct” button then the concatenation of first character of str1,the first character of str2, second character of str1, second character of str2 and so on.., is performed to create the new string. The remaining characters at the end of str1or str2 are concatenated to the end of the resultant string. | | |

# PART-A

**Program 01: Create C# console application which reads a set of strings and then prints the string having maximum number of vowels.**

**SOURCE CODE:**

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace Vowels
{
   class Program
   {
      static void Main(string[] args)
      {
         Console.Write("How many strings do you want to enter: ");
         int n = int.Parse(Console.ReadLine());
         string maxVowelString = "";
         int maxVowels = 0;
         for (int i = 1; i <= n; i++)
         {
            Console.Write("Enter string " + i + ": ");
            string input = Console.ReadLine();
            int vowelCount = 0;
            foreach (char c in input)
            {
               if ("aeiouAEIOU".IndexOf(c) >= 0)
               {
                  vowelCount++;
               }
            }
            if (vowelCount > maxVowels)
            {
               maxVowels = vowelCount;
               maxVowelString = input;
            }
         }
         Console.WriteLine("\nString with most vowels: " + maxVowelString);
         Console.WriteLine("Number of vowels: " + maxVowels);
         Console.ReadKey();
      }
   }
}
```

**OUTPUT**

```
How many strings do you want to Enter: 3
Enter string 1: Adithya
Enter string 2: Abhishek
Enter string 3: Bharathesh

String with most vowels: Adithya
Number of vowels: 3
```

**Program 02: Write C# console program which stores different types of topics with the values Topics = {"Introduction to C#", "Variables", "Data Types", "Loops", "If statements", "jump statements", "classes and objects", "Inheritance", "Constructors"} Using switch case statement categorise the topics as "Basic", "Control Flow" and "OOPs Concepts".**

**SOURCE CODE:**

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;


namespace list
{
   class Program
   {
      static void Main(string[] args)
      {
         List<string> Topics = new List<string>()
          {
            "Introduction to C#",
            "Variables",
            "Data Types",
             "Loops",
               "If statements",
             "Jump Statements",
             "Class & Object",
             "Inheritance",
             "Constructors"
             };
         foreach (string topic in Topics)
         {
```

```
            string category;
            switch (topic)
            {
                case "Introduction to C#":
                case "Variables":
                case "Data Types":
                    category = "Basic";
                    break;
                case "Loops":
                case "If statements":
                case "Jump Statements":
                    category = "Control Flow";
                    break;
                case "Class & Object":
                case "Inheritance":
                case "Constructors":
                    category = "OOPs Concepts";
                    break;
                default:
                    category = "Unknown";
                    break;
            }
            Console.WriteLine("Topic is " + topic + "; Category is " + category);
        }
        Console.ReadKey();
    }
  }
}
```

**OUTPUT**

```
Topic is Introduction to C#; Category is Basic
Topic is Variables; Category is Basic
Topic is Data Types; Category is Basic
Topic is Loops; Category is Control Flow
Topic is If statements; Category is Control Flow
Topic is Jump Statements; Category is Control Flow
Topic is Class & Object; Category is OOPs Concepts
Topic is Inheritance; Category is OOPs Concepts
Topic is Constructors; Category is OOPs Concepts
```

**Program 03: Write a console application to generate a number pattern in the shape of a right angled triangle with the numbers increased by 1.**

**SOURCE CODE**

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace Pattern
{
    internal class Program
    {
        static void Main(string[] args)
        {
            int rows = 4;
            int num = 1;
            for (int i = 1; i <= rows; i++)
            {
                for (int j = 1; j <= i; j++)
                {
                    Console.Write(num + " ");
                    num++;
                }
                Console.WriteLine();
            }
                Console.ReadKey();
        }
    }
}
```

**OUTPUT**

```
1
2 3
4 5 6
7 8 9 10
```

**Program 04: Create a C# console application that receives the following information from a set of students: Student Id, Student Name, Course Name, and Date of Birth. The application should also display the information of all the students once the data is entered. Implement this using an array of objects.**

**SOURCE CODE :**

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace Students1
{
   class Student
   {
      public int Id;
      public string Name;
      public string Course;
      public DateTime DOB;
   }
   class Program
   {
      static void Main(string[] args)
      {
         Console.Write("Enter number of students: ");
         int n = int.Parse(Console.ReadLine());

         Student[] students = new Student[n];

         for (int i = 0; i < n; i++)
         {
            students[i] = new Student();
            Console.WriteLine("\nStudent " + (i + 1));

            Console.Write("ID: ");
            students[i].Id = int.Parse(Console.ReadLine());

            Console.Write("Name: ");
            students[i].Name = Console.ReadLine();

            Console.Write("Course: ");
            students[i].Course = Console.ReadLine();
```

```
            Console.Write("Date of Birth (yyyy-mm-dd): ");
            students[i].DOB = DateTime.Parse(Console.ReadLine());
        }

        Console.WriteLine("\n--- Student Details ---");
        for (int i = 0; i < n; i++)
        {
            Console.WriteLine("ID: " + students[i].Id +
                              ", Name: " + students[i].Name +
                              ", Course: " + students[i].Course +
                              ", DOB: " + students[i].DOB.ToString("dd-MM-yyyy"));
        }
    }
  }
}
```

**OUTPUT:**

```
Enter number of Students :3

 Student1
ID:101
Name:sandeep
Course:BCA
Date of Birth(yyyy-mm-dd):2006-09-26

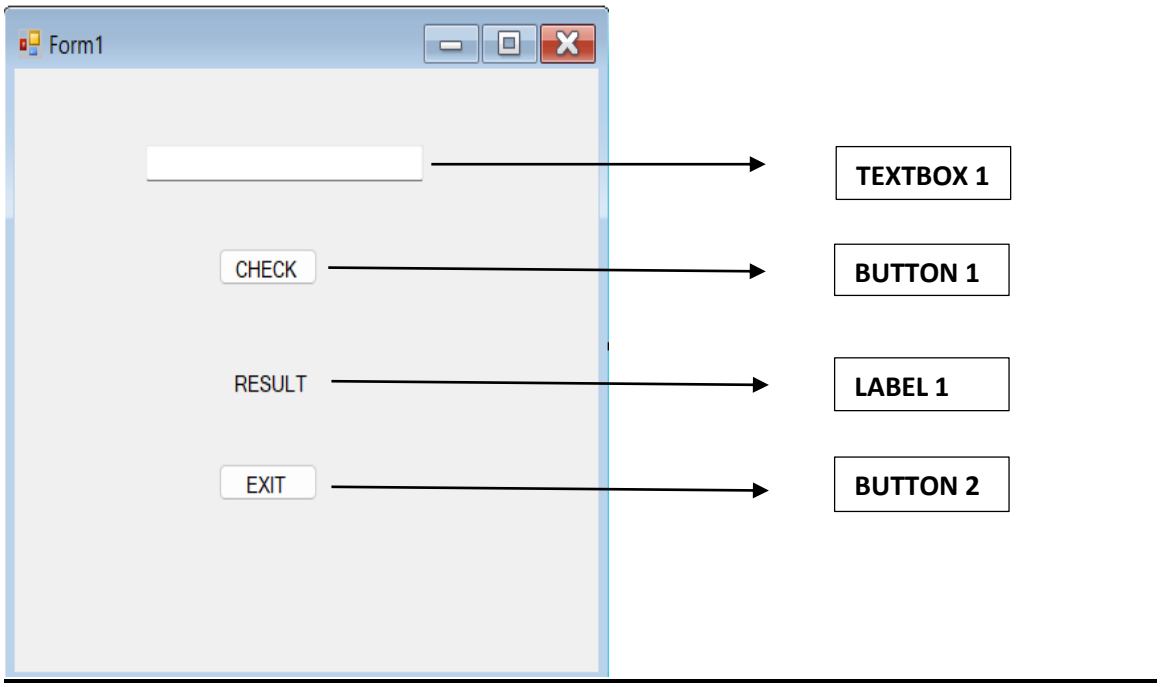
 Student2
ID:102
Name:sudarshan
Course:BCA
Date of Birth(yyyy-mm-dd):2006-12-22

 Student3
ID:103
Name:vikas
Course:BCA
Date of Birth(yyyy-mm-dd):2007-05-20

---Student Details---
Id: 101
Name: sandeep
Course: BCA
DOB: 26-09-2006
Id: 102
Name: sudarshan
Course: BCA
DOB: 22-12-2006
Id: 103
Name: vikas
Course: BCA
DOB: 20-05-2007
```

**Program 05: Create a windows form application that allows the user to enter a number in the textbox named 'getnum'. Check whether the number in the textbox 'getnum' is palindrome or not. Print the message accordingly in the label control named lbldisplay when the user clicks on the button 'check'.**

**DESIGN :**



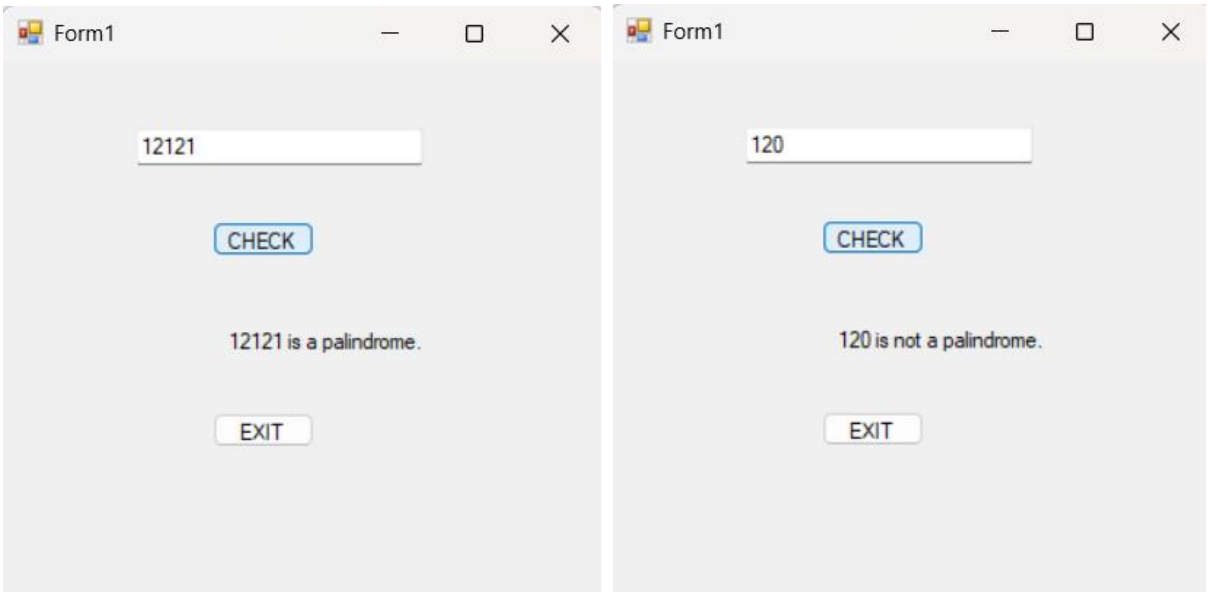

| SL.NO | CONTROLS | NAME | TEXT |
|---|---|---|---|
| 1 | Textbox 1 | getnum | |
| 2 | Button 1 | check | CHECK |
| 3 | Label 1 | lbldisplay | RESULT |
| 4 | Button 2 | btnexit | EXIT |

**SOURCE CODE :**

```
Public Class Form1
    Private Sub check_Click(ByVal sender As System.Object, ByVal e As System.EventArgs)
Handles check.Click
        Dim input As String = getnum.Text.Trim()
        If input = " " Then
            lbldisplay.Text = "Please enter a number."
            Exit Sub
        End If
        Dim num As Long
        If Not Long.TryParse(input, num) Then
            lbldisplay.Text = "Please enter a valid number."
```

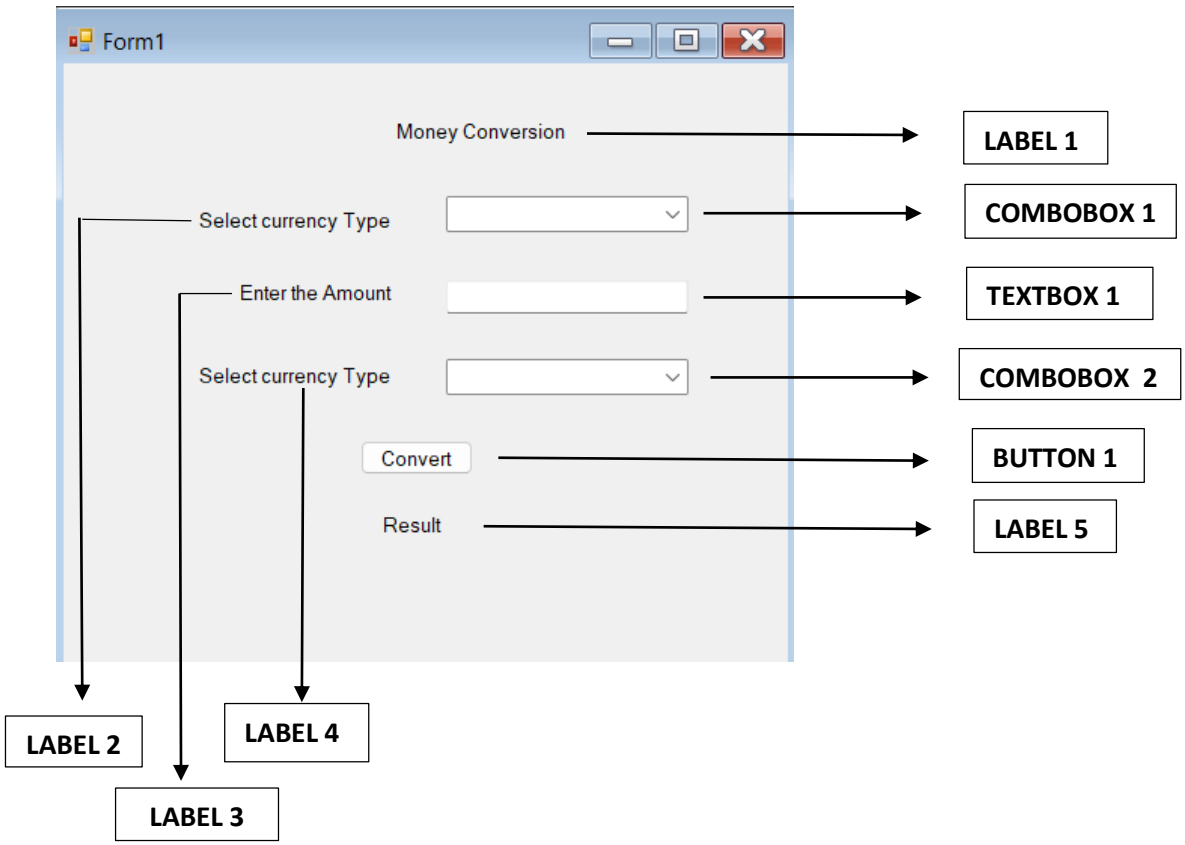
```
            Exit Sub
        End If
        Dim reversed As String = StrReverse(input)
        If input = reversed Then
            lbldisplay.Text = input & " is a palindrome."
        Else
            lbldisplay.Text = input & " is not a palindrome."
        End If
    End Sub
Private Sub btnexit_Click(ByVal sender As System.Object, ByVal e As System.EventArgs)
Handles btnexit.Click
        End
    End Sub
End Class
```

**OUTPUT :**



**Program 06: Implement C# windows form application to perform money conversion.**

**DESIGN:**




| SL.NO | CONTROLS | NAME | TEXT |
|---|---|---|---|
| 1 | Label 1 | lblTitle | Money Conversion |
| 2 | Label 2 | lblFrom | Select currency Type |
| 3 | Label 3 | lblAmount | Enter the Amount |
| 4 | Label 4 | lblTo | Select currency Type |
| 5 | Label 5 | lblResult | Result |
| 6 | Combobox 1 | cmbFrom | |
| 7 | Combobox 2 | cmbTo | |
| 8 | Textbox 1 | txtAmount | |
| 9 | Button 1 | btnConvert | Convert |

**SOURCE CODE :**

```
using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Drawing;
using System.Linq;
using System.Threading.Tasks;
using System.Windows.Forms;
```

```
namespace conversion
{
   public partial class Form1 : Form
   {
      public Form1()
      {
         InitializeComponent();
      }
      private void btnConvert_Click(object sender, EventArgs e)
      {
         {
            try
            {
               decimal amount = Convert.ToDecimal(txtAmount.Text);
               string fromCurrency = cmbFrom.SelectedItem.ToString();
               string toCurrency = cmbTo.SelectedItem.ToString();

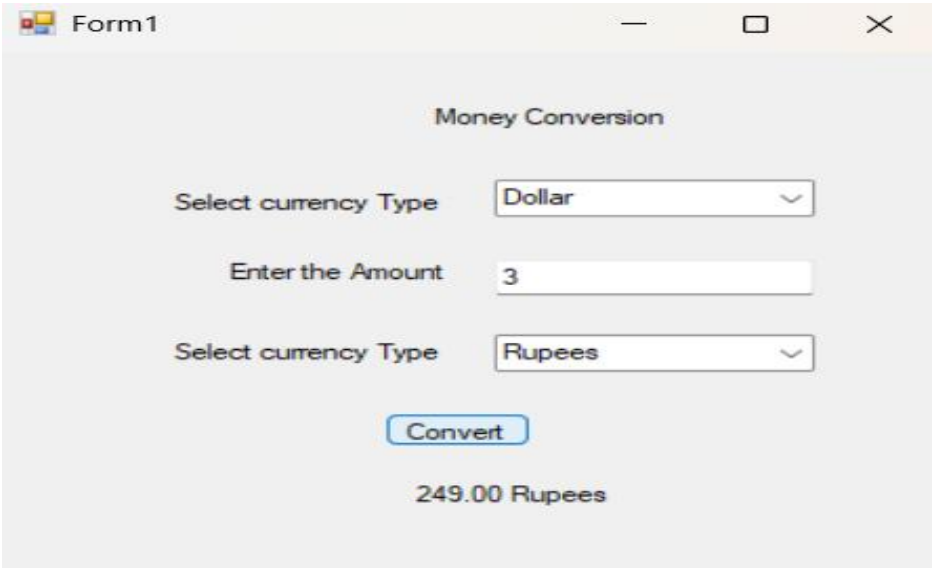
               decimal convertedAmount = ConvertCurrency(amount, fromCurrency,
toCurrency);

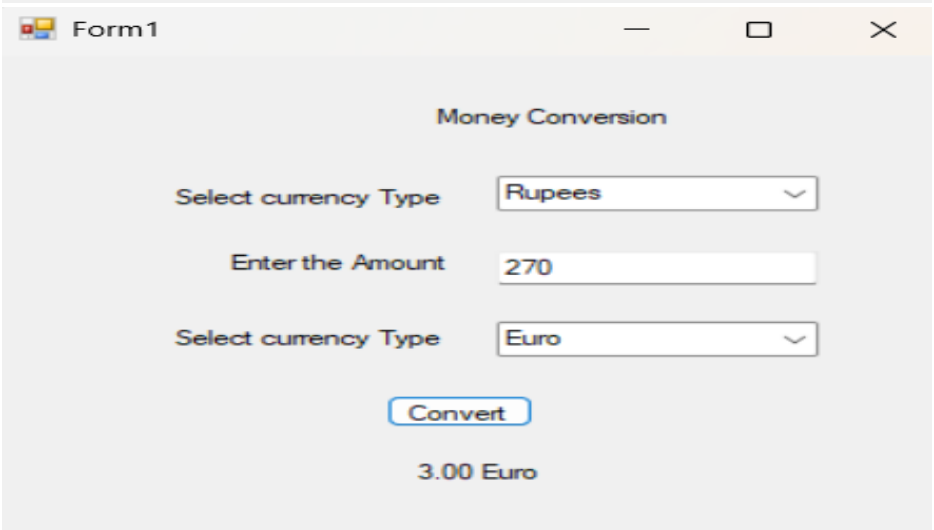
               lblResult.Text = convertedAmount.ToString("F2") + " " + toCurrency;
            }
            catch (Exception ex)
            {
               MessageBox.Show("Error: " + ex.Message, "Conversion Failed",
                  MessageBoxButtons.OK, MessageBoxIcon.Error);
            }
         }
      }
      private decimal ConvertCurrency(decimal amount, string fromCurrency, string
toCurrency)
      {
         decimal inrValue = 0;

         // Convert FROM currency → INR
         if (fromCurrency == "Dollar")
            inrValue = amount * 83;
         else if (fromCurrency == "Euro")
            inrValue = amount * 90;
         else if (fromCurrency == "Diram")
            inrValue = amount * 21;
         else if (fromCurrency == "Rupees")
          inrValue = amount;
```

```
            // Convert INR → TO currency
            decimal result = 0;
            if (toCurrency == "Dollar")
                result = inrValue / 83;
            else if (toCurrency == "Euro")
                result = inrValue / 90;
            else if (toCurrency == "Diram")
                result = inrValue / 21;
            else if (toCurrency == "Rupees")
                result = inrValue;
            return result;
        }
    }
}
```

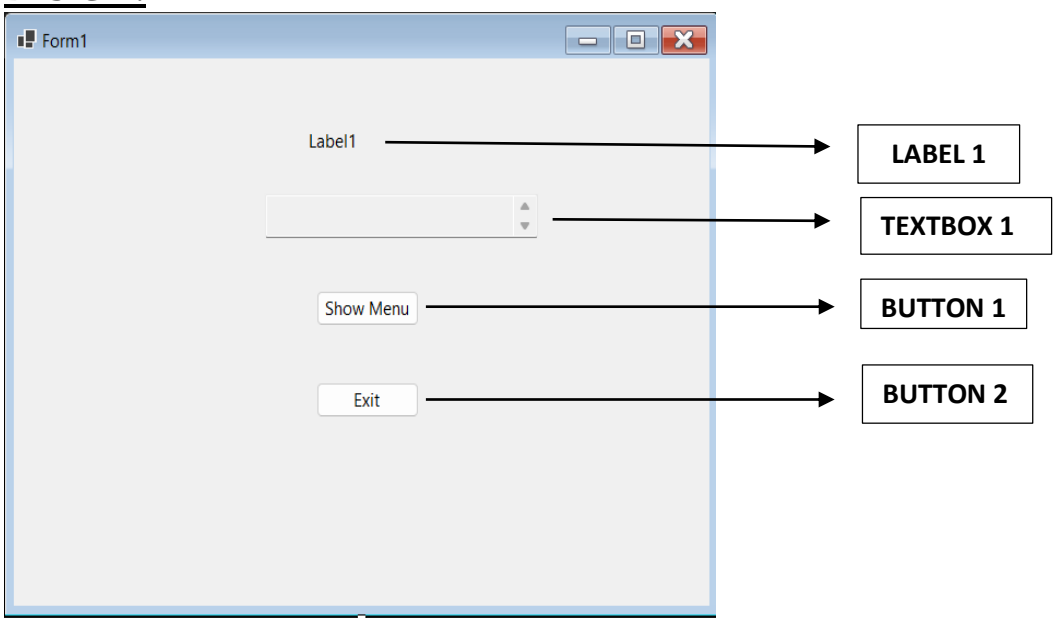
**OUTPUT :**



**Program 07: Design a webpage of a hotel which displays different Menu as per the Time of Visit.**

**DESIGN :**




| SL.NO | CONTROLS | NAME | TEXT |
|---|---|---|---|
| 1 | Label 1 | lblGreeting | Label1 |
| 2 | Textbox 1 | txtMenu | |
| 3 | Button 1 | btnShowMenu | Show Menu |
| 4 | Button 2 | btnexit | Exit |

**SOURCE CODE :**

```
Public Class Form1
    Private Sub btnShowMenu_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles btnShowMenu.Click

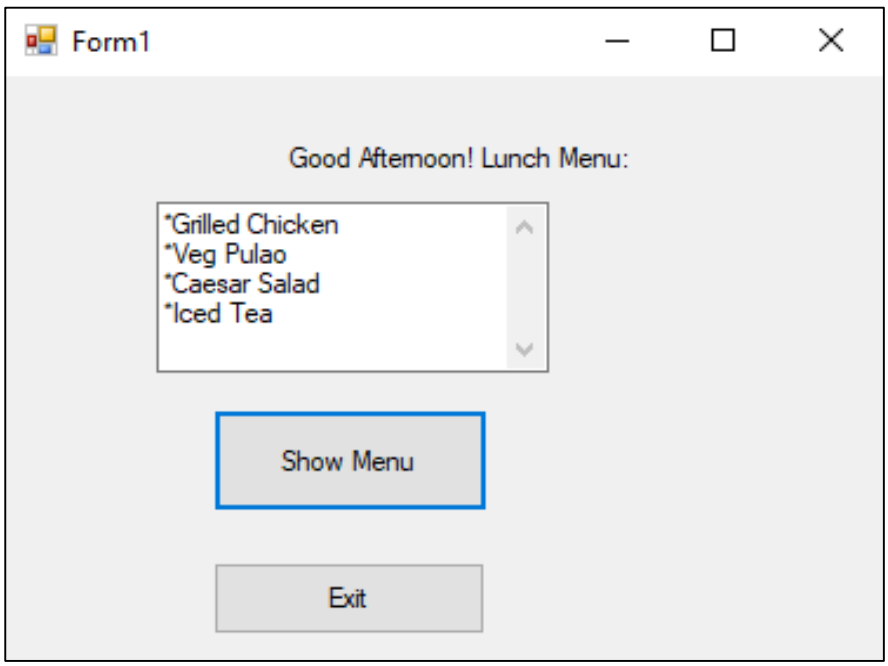
        Dim hour As Integer = DateTime.Now.Hour
        Dim menuItems As String = ""
        If hour >= 6 AndAlso hour < 11 Then
            lblGreeting.Text = "Good Morning! Breakfast Menu:"
            menuItems = "*pancakes with Honey" & vbCrLf &
                        "*Omelette" & vbCrLf &
                        "*Fresh Juice" & vbCrLf &
                        "*Coffee"
        ElseIf hour >= 11 AndAlso hour < 17 Then
            lblGreeting.Text = "Good Afternoon! Lunch Menu:"
            menuItems = "*Grilled Chicken" & vbCrLf &
                        "*Veg Pulao" & vbCrLf &
```

```
                "*Caesar Salad" & vbCrLf &
                "*Iced Tea"
        ElseIf hour >= 17 AndAlso hour < 20 Then
            lblGreeting.Text = "Good Evening! Dinner Menu:"
            menuItems = "*Paneer Butter Masala" & vbCrLf &
                "*Naan" & vbCrLf &
                "*Biryani" & vbCrLf &
                "*Gulab Jamun"
        Else

            lblGreeting.Text = " Late Night Snacks Menu:"
            menuItems = "*SandWich" & vbCrLf &
                "*French Fries" & vbCrLf &
                "*Soup" & vbCrLf &
                "*Hot Chocolate"
        End If
        txtMenu.Text = menuItems
    End Sub

    Private Sub btnexit_Click (ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles btnexit.Click
        End
    End Sub
End Class
```

**OUTPUT:**

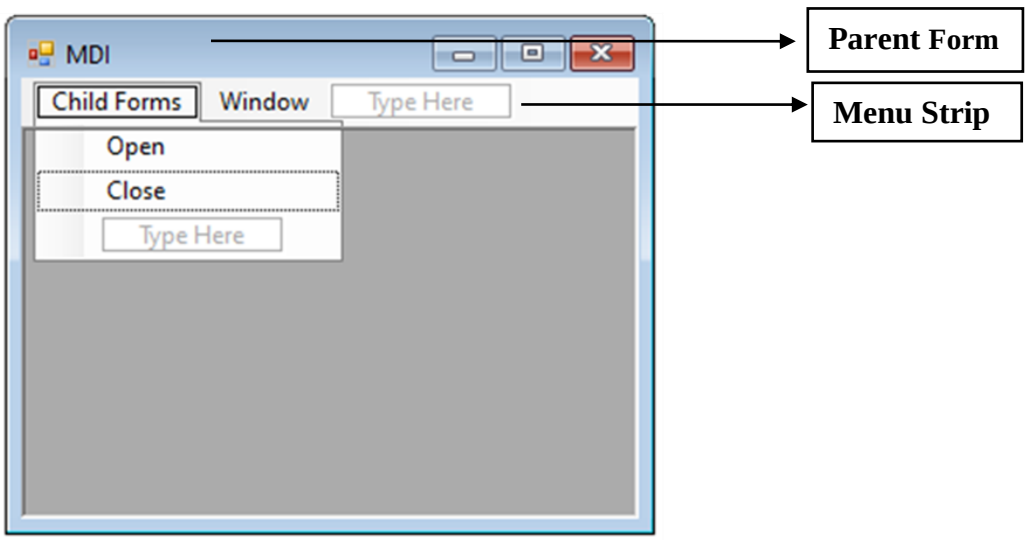


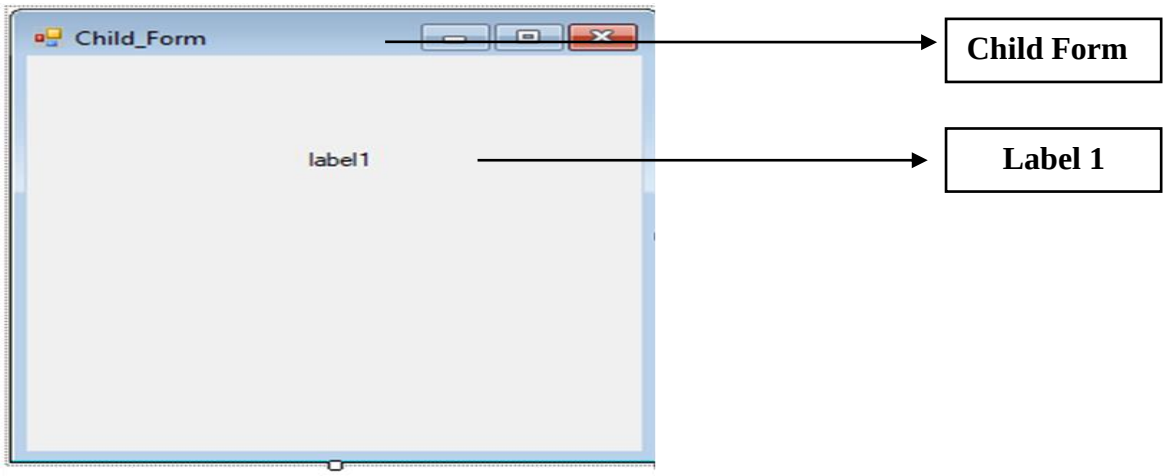
# PART-B

**Program 01: Create an MDI form with several child forms. (at least 3 child forms with proper information) MDI parent form should contain menu strip as shown below with menu options to open, close and rearrange the child forms.**

| Child Forms | Window |
|---|---|
| Open | Cascade |
| Close | Tile Horizontal |
| | Tile Vertical |
| | Arrange icons. |

**DESIGN:**




| SL NO | CONTROLS | NAME | TEXT |
|---|---|---|---|
| 1 | MENUSTRIP | ----- | ----- |
| 2 | LABEL1 | Child | CHILD FORM |

**SOURCE CODE:**

**Parent Form Code:**

```
using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Drawing;
using System.Linq;
using System.Text;
using System.Windows.Forms;

namespace MDI_Form
{
   public partial class Form1 : Form
   {
      long count = 0;
      public Form1()
      {
         InitializeComponent();
      }

      private void openToolStripMenuItem_Click(object sender, EventArgs e)
      {
         if (this.MdiChildren.Length == 0)
            count = 0;
         else
            count = this.MdiChildren.Length;
            Child_Form f = new Child_Form(++count);
         f.MdiParent = this;
         f.Show();
      }

      private void closeToolStripMenuItem_Click(object sender, EventArgs e)
      {
         if (this.ActiveMdiChild != null)
            this.ActiveMdiChild.Close();
      }

      private void cascadeToolStripMenuItem_Click(object sender, EventArgs e)
      {
         this.LayoutMdi(MdiLayout.Cascade);
      }

      private void tileHorizontalToolStripMenuItem_Click(object sender, EventArgs e)
      {
```

```
            this.LayoutMdi(MdiLayout.TileHorizontal);
        }

        private void tileVerticalToolStripMenuItem_Click(object sender, EventArgs e)
        {
            this.LayoutMdi(MdiLayout.TileVertical);
        }
             private void arrangeIconsToolStripMenuItem_Click(object sender, EventArgs e)
        {
            this.LayoutMdi(MdiLayout.ArrangeIcons);
        }
    }
}
```

**Child Form Code:**

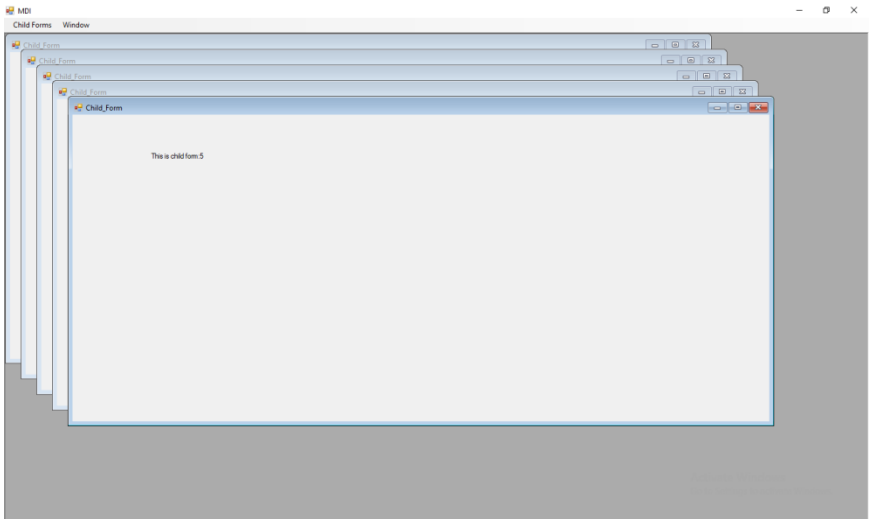
```
using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Drawing;
using System.Linq;
using System.Text;
using System.Windows.Forms;

namespace MDI_Form
{
    public partial class Child_Form : Form
    {
        public Child_Form(long count)
        {
            InitializeComponent();
            Child.Text = "This is child form:" + count;
        }

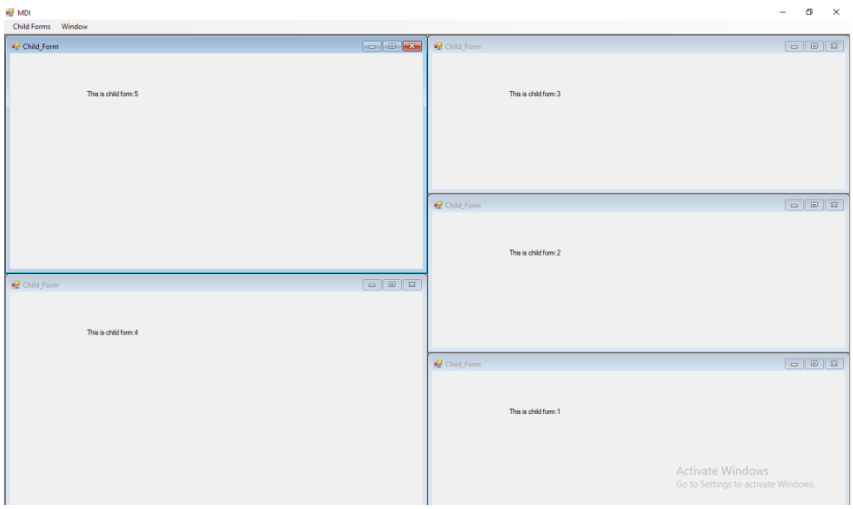
        private void Child_Form_Load(object sender, EventArgs e)
        {
        }
    }
}
```

**OUTPUT:**

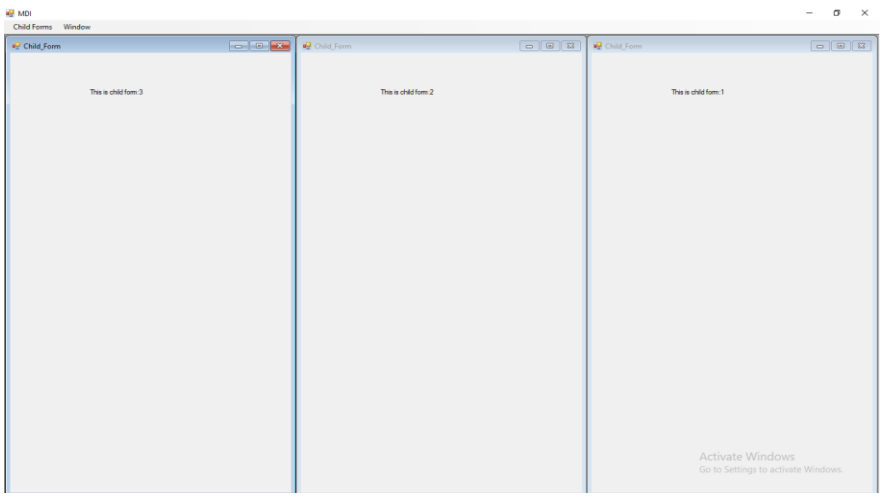
## Cascade




## Tile Vertical:




## Tile Horizontal:



**Program 02: Design and implement a windows form application in C# to calculate and display an employee's salary details using the given conditions.**

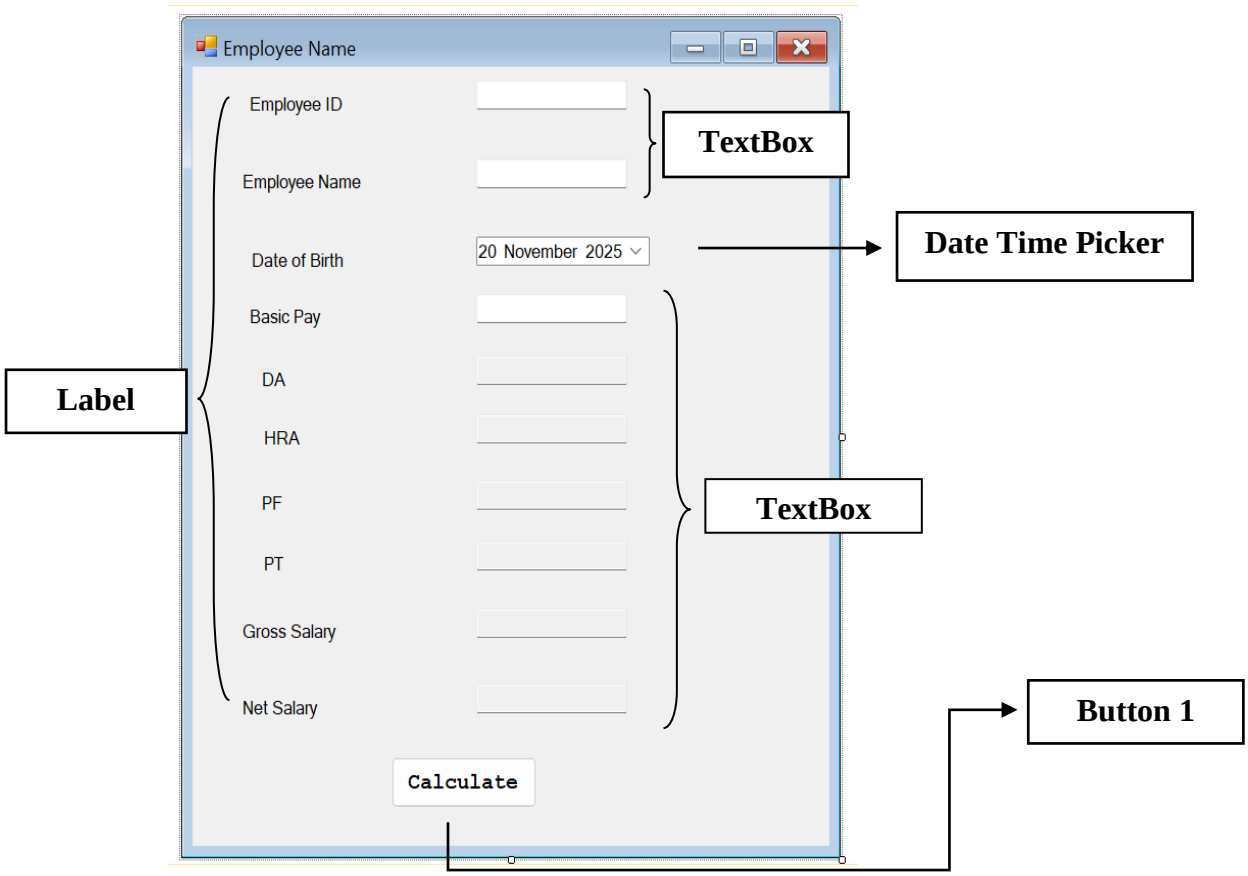
**If Basic <= 20000 D.A is 40% of Basic, H.R.A is 10% of Basic, P.F 12% of Gross, PT is Rs.100**

**If Basic> 20000 D.A is 50% of Basic, H.R.A 15% of Basic, P.F 12% of Gross, PT is Rs.150**

**Gross = Basic+D.A +HRA, Net = Gross -PT –PF**

**The form should allow the user to input Employee ID, Employee Name, Date of Birth and Basic Pay using appropriate controls (Textbox/DateTimePicker). On clicking "Calculate" button, the application should display the calculated values in appropriate output controls (ReadOnlyTextBoxes or Labels).**

**DESIGN:**




| SL NO | CONTROLS | NAME | TEXT |
|---|---|---|---|
| 1 | Label1 | Label1 | Employee ID |
| 2 | Label2 | Label2 | Employee Name |
| 3 | Label3 | Label3 | Date of Birth |
| 4 | Label4 | Label4 | Basic Pay |
| 5 | Label5 | Label5 | DA |
| 6 | Label6 | Label6 | HRA |
| 7 | Label7 | Label7 | PF |
| 8 | Label8 | Label8 | PT |

| 9 | Label9 | Label9 | Gross Salary |
|---|---|---|---|
| 10 | Label10 | Label10 | Net Salary |
| 11 | TextBox1 | txtEmpID | ----- |
| 12 | TextBox2 | txtEmpName | ----- |
| 13 | DateTimePicker | dtDOB | ----- |
| 14 | TextBox3 | txtBasic | ----- |
| 15 | TextBox4 | txtDA | ----- |
| 16 | TextBox5 | txtHRA | ----- |
| 17 | TextBox6 | txtPF | ----- |
| 18 | TextBox7 | txtPT | ----- |
| 19 | TextBox8 | txtGross | ----- |
| 20 | TextBox9 | txtNet | ----- |
| 21 | Button | btnCalculate | Calculate |

**SORCE CODE:**

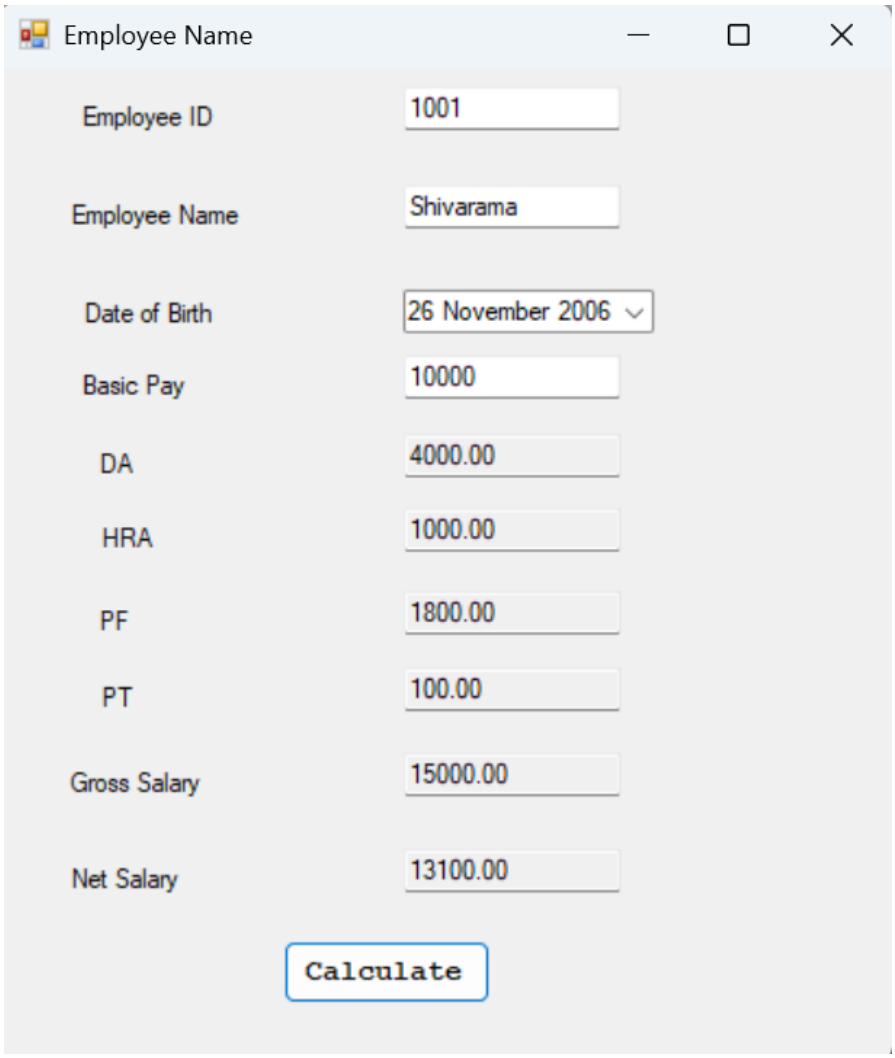
```
using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Drawing;
using System.Linq;
using System.Text;
using System.Windows.Forms;

namespace EmployeeSalary
{
   public partial class Form1 : Form
   {
      public Form1()
      {
         InitializeComponent();
      }

      private void btnCalculate_Click(object sender, EventArgs e)
      {
         try
         {
            string empID = txtEmpID.Text;
            string empName = txtEmpName.Text;
            DateTime dob = dtpDOB.Value;
            double basic = Convert.ToDouble(txtBasic.Text);
            double da, hra, pf, pt, gross, net;
            if (basic <= 20000)
            {
```

```
                da = 0.40 * basic;
                hra = 0.10 * basic;
                pt = 100;
            }
            else
            {
                da = 0.50 * basic;
                hra = 0.15 * basic;
                pt = 150;
            }
            gross = basic + da + hra;
            pf = 0.12 * gross;
            net = gross - (pt + pf);
            txtDA.Text = da.ToString("F2");
            txtHRA.Text = hra.ToString("F2");
            txtPF.Text = pf.ToString("F2");
            txtPT.Text = pt.ToString("F2");
            txtGross.Text = gross.ToString("F2");
            txtNet.Text = net.ToString("F2");
        }
        catch (Exception ex)
        {
            MessageBox.Show("Please enter valid input." + ex.Message);
        }
    }
  }
}
```

**OUTPUT:**

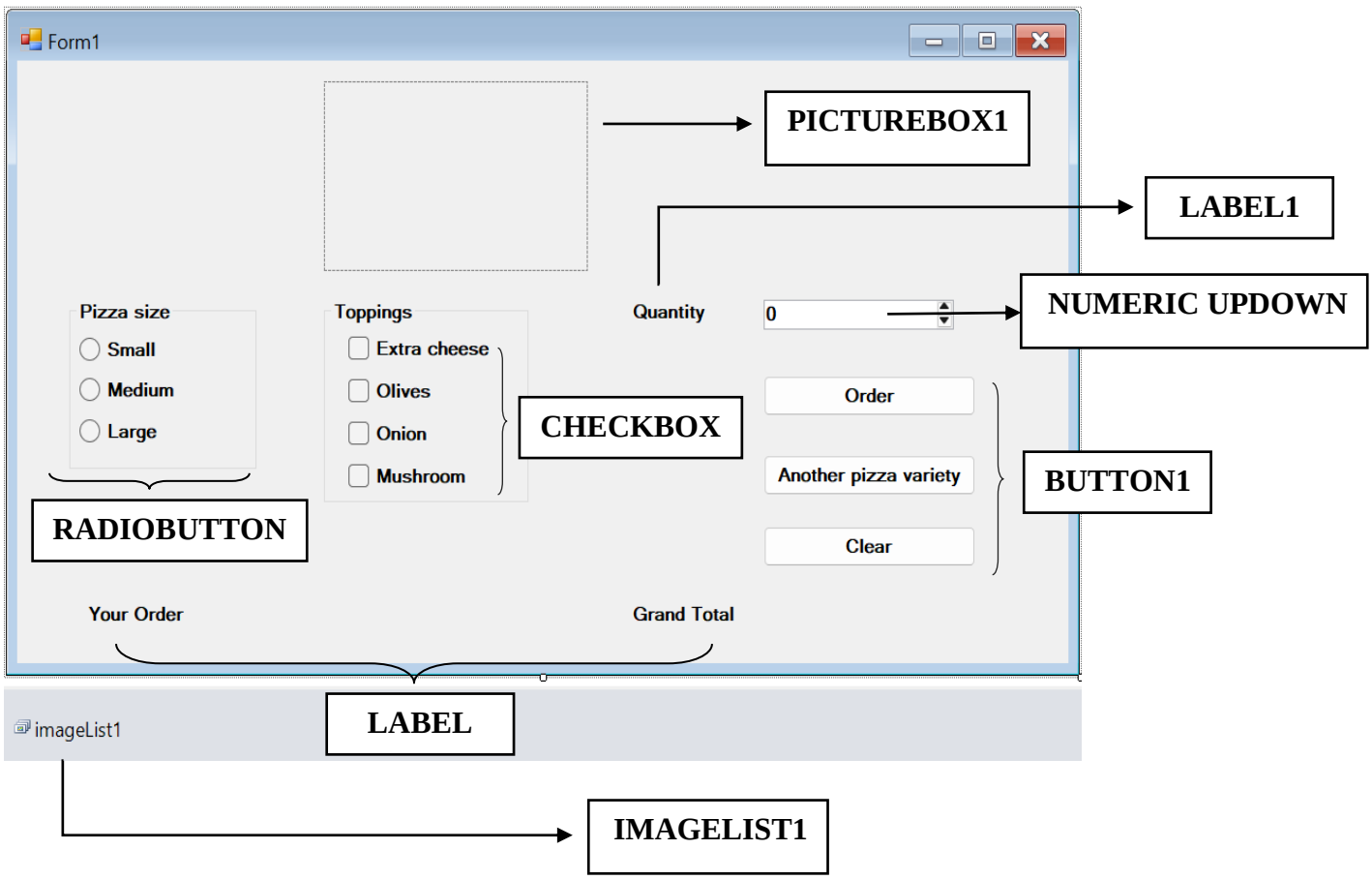

Employee Name
Employee ID
1001
Employee Name
Shivarama
Date of Birth
26 November 2006
Basic Pay
10000
DA
4000.00
HRA
1000.00
PF
1800.00
PT
100.00
Gross Salary
15000.00
Net Salary
13100.00
Calculate

**Program 03: Design windows form application to order pizza in a restaurant. Select pizza size using radio buttons, toppings using check boxes and quantity using numericUpDown control. PictureBox should display an image of the pizza with selected toppings.**

**Buttons : i) Order – To calculate and display the total amount and order summary. ii)Another pizza variety – Clears earlier size, toppings and quantity selections in the form so that you can add another variety of pizza to the current order. iii)Clear : cancels the order.**

| Pizza Size Selection using RadioButtons | Toppings Selection using CheckBoxes |
|---|---|
| ○ Small – ₹100<br>○ Medium – ₹150<br>○ Large – ₹200<br>*(Note: Only one size can be selected.)* | ○ Extra cheese – ₹30<br>○ Olives – ₹30<br>○ Onions – ₹20<br>○ Mushrooms – ₹30 |

**DESIGN:**




| SL NO | CONTROLS | NAME | TEXT |
|---|---|---|---|
| 1 | GroupBox 1 | grpSize | Pizza size |
| 2 | RadioButton1 | rdbSmall | Small ₹100 |
| 3 | RadioButton2 | rdbMedium | Medium ₹150 |
| 4 | RadioButton3 | rdbLarge | Large ₹200 |
| 5 | GroupBox2 | grpToppings | Toppings |

| | | | |
|---|---|---|---|
| **6** | **CheckBox1** | **ChkCheese** | **Extra cheese ₹30** |
| **7** | **CheckBox2** | **ChkOlives** | **Olives ₹30** |
| **8** | **CheckBox3** | **ChkOnion** | **Onion ₹20** |
| **9** | **CheckBox4** | **ChkMushroom** | **Mushroom ₹30** |
| **10** | **Numeric UpDown** | **numQuantity** | **-----** |
| **11** | **ImageList** | **imageList1** | **-----** |
| **12** | **PictureBox** | **pictureBox1** | **-----** |
| **13** | **Button1** | **btnOrder** | **Order** |
| **14** | **Button2** | **btn** | **Another pizza variety** |
| **15** | **Button3** | **btnClear** | **Clear** |
| **16** | **Label1** | **lblTotal** | **Grand Total** |
| **17** | **Label2** | **lblOrder** | **Your Order** |

**SOURCE CODE:**

```
using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Drawing;
using System.Linq;
using System.Text;
using System.Windows.Forms;

namespace  pizzaoder
{
   public partial class Form1 : Form
   {
      int grandTotal = 0;
      int orderCount = 0;
      public Form1()
      {
         InitializeComponent();
      }

      private void Form1_Load(object sender, EventArgs e)
      {
         lblOrder.Text = "Your order:\n";
         lblAmount.Text = "Grand Total:Rs.0";
         if (imageList1.Images.Count > 0)
         {
```

```
        pictureBox1.SizeMode = PictureBoxSizeMode.StretchImage;
        pictureBox1.Image = imageList1.Images[0];
    }

}
private void UpdatePizzaImage()
{
    if (rdbSmall.Checked) pictureBox1.Image = imageList1.Images[0];
    else if (rdbMedium.Checked) pictureBox1.Image = imageList1.Images[1];
    else if (rdbLarge.Checked) pictureBox1.Image = imageList1.Images[2];
}

private void rdbSmall_CheckedChanged(object sender, EventArgs e)
{
    UpdatePizzaImage();
}

private void rdbMedium_CheckedChanged(object sender, EventArgs e)
{
    UpdatePizzaImage();
}

private void rdbLarge_CheckedChanged(object sender, EventArgs e)
{
    UpdatePizzaImage();
}

private void btnOrder_Click(object sender, EventArgs e)
{
    int price = 0;
    string size = "";
    if (rdbSmall.Checked)
    {
        price = 100; size = "small";
    }
    else if (rdbMedium.Checked)
    {
        price = 150; size = "Meduim";
    }
  else if(rdbLarge.Checked)
    {
        price = 200; size = "Large";


  }
    else{
```

```
        MessageBox.Show("please select a pizza size!");
        return ;

    }
    List<string >toppings=new List<string >();
    if(chkCheese.Checked )
    {
        toppings.Add("extra Cheese");
        price+=30;
    }
    if(chkOlives.Checked )
    {
        toppings.Add("Olives");
        price+=30;
    }
    if(chkOnion.Checked )
    {
        toppings.Add("Onion");
        price+=20;
    }
    if(chkMushroom.Checked )
    {
        toppings.Add("mushroom");
        price+=30;
    }
    string toppingsText=toppings .Count >0? string.Join (",",toppings ):"no toppings";
    int qty=(int)numQuantity.Value ;
    int total=price *qty;
    orderCount ++;
    grandTotal +=total ;
    lblOrder.Text +="\n" + orderCount +"." + qty +" " + size +"sized pizza with" +
    toppingsText +"->Rs" + total ;
    lblAmount.Text="Grand Total :"+grandTotal;

        UpdatePizzaImage();
}

private void btnAnother_Click(object sender, EventArgs e)
{
    rdbSmall.Checked = false;
    rdbMedium.Checked = false;
    rdbLarge.Checked = false;
    chkCheese.Checked = false;
    chkOlives.Checked = false;
    chkOnion.Checked = false;
    chkMushroom.Checked = false;
```

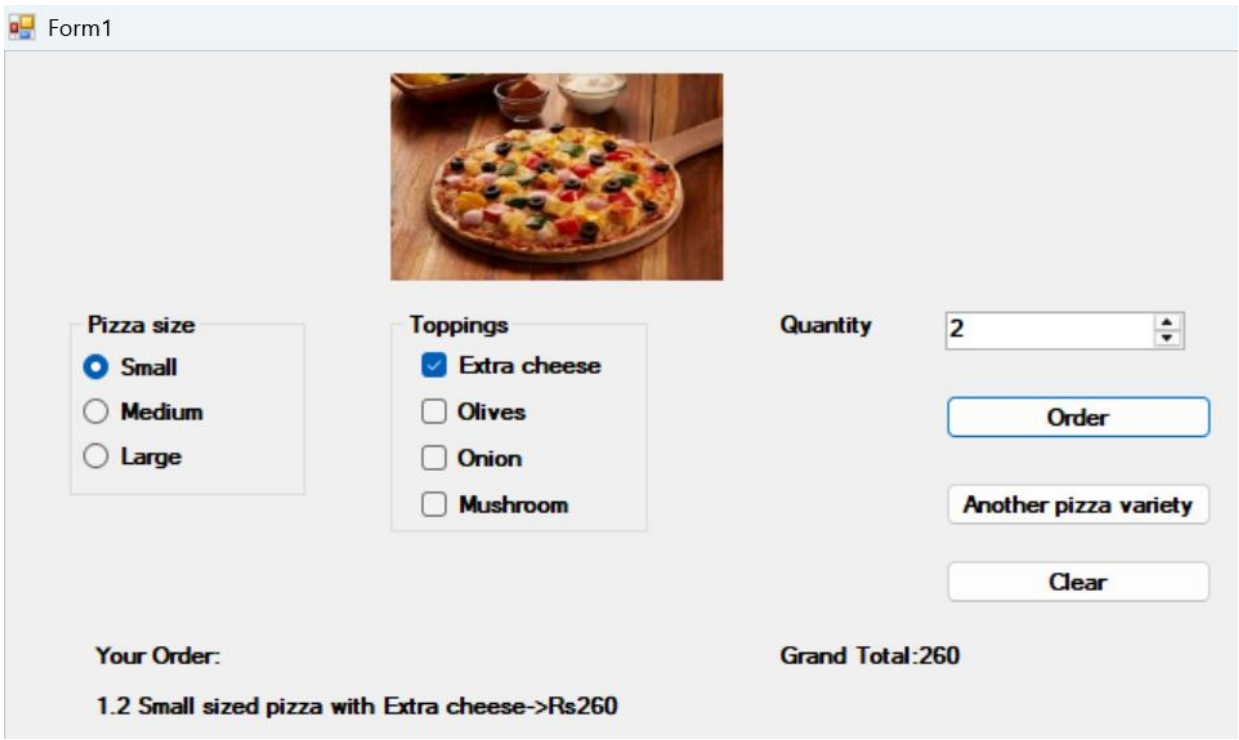
```
            numQuantity.Value = 1;
            pictureBox1.Image = null;

        }

        private void btnclear_Click(object sender, EventArgs e)
        {
            lblOrder.Text = "Your order:\n";
            lblAmount.Text = "Grand Total:Rs";
            pictureBox1.Image = null;
            rdbSmall.Checked = false;
            rdbMedium.Checked = false;
            rdbLarge.Checked = false;
            chkCheese.Checked = false;
            chkOlives.Checked = false;
            chkOnion.Checked = false;
            chkMushroom.Checked = false;
            numQuantity.Value = 1;
            grandTotal = 0;
            orderCount = 0;
        }

    }
}
```

**OUTPUT:**



**Program 04: Create a web form for Login Module which adds a username and password to the database. The username in the database should be a primary key. Also implement proper authentication functionality.**

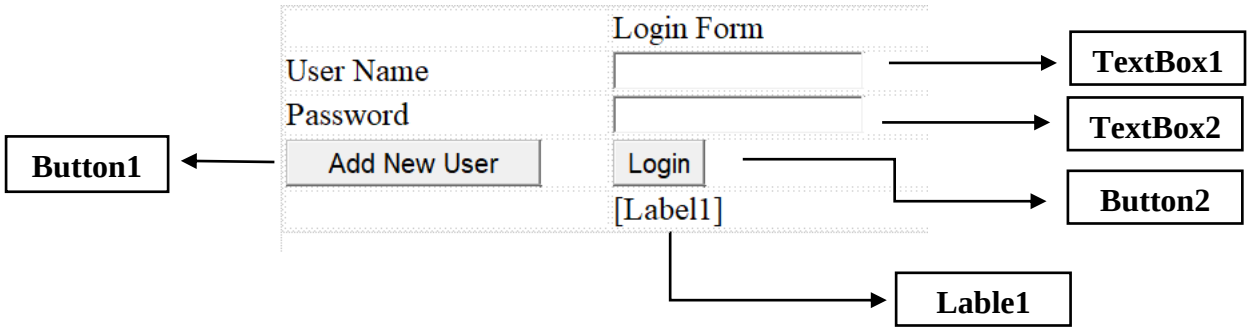
### Creating a Web Application:-

Click on the icon of Visual Studio 2010 → New Project → Visual C# → Web → ASP.NET Empty Web Application → Name the Web Application → Browse the location→ Ok.

### Creating a Web Form:

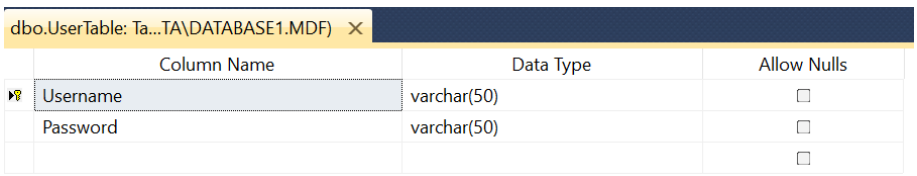
Right click on the name of the Web Application → Add → New Item → Visual C# Web →Web Form → Add

### DESIGN:



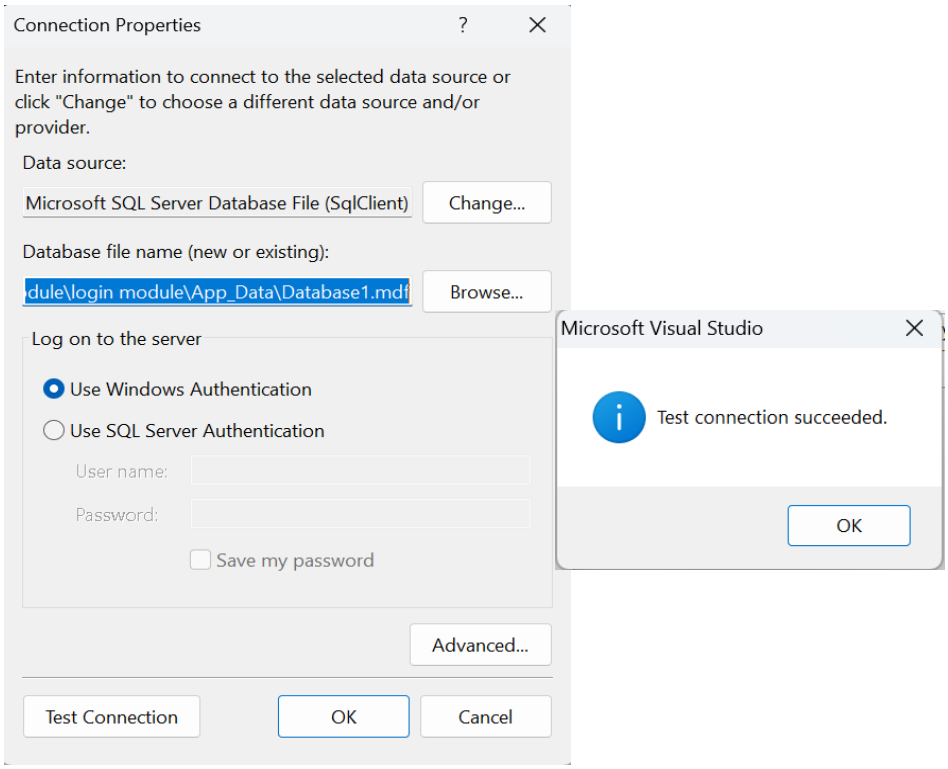

| SL NO | CONTROLS | NAME | TEXT |
|---|---|---|---|
| 1 | TextBox1 | TextBox1 | ----- |
| 2 | TextBox2 | TextBox2 | ----- |
| 3 | Button1 | Button1 | Add new user |
| 4 | Button2 | Button2 | Login |
| 5 | Lable1 | Label1 | ----- |

### Database:




### Connection:

Under Solution Explorer → Properties → Settings → Browse Microsoft SQL Server Database File (SqlClient) →Test Connection → Ok.




**SOURCE CODE:**

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;
using System.Data.SqlClient;

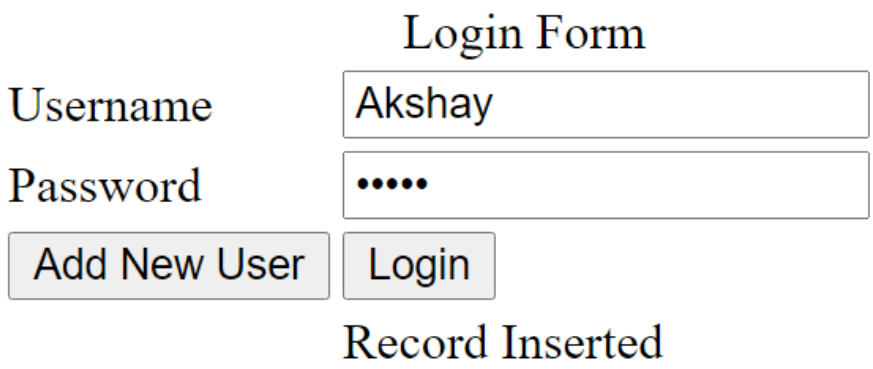
namespace Login_module
{
   public partial class WebForm1 : System.Web.UI.Page
   {
      protected void Button1_Click(object sender, EventArgs e)
      {
         SqlConnection con = new SqlConnection(Properties.Settings.Default.constr);
         string sql = "insert into UserTable values(@uname,@passwd)";
         SqlCommand cmd = new SqlCommand(sql,con);
         cmd.Parameters.AddWithValue("@uname", TextBox1.Text);
         cmd.Parameters.AddWithValue("@passwd", TextBox2.Text);
         string sqlchk = "select count(*) from UserTable where Username=@chkname";
         SqlCommand cmdchk = new SqlCommand(sqlchk, con);
         cmdchk.Parameters.AddWithValue("@chkname", TextBox1.Text);
         con.Open();
         SqlDataReader rdr = cmdchk.ExecuteReader();
         rdr.Read();
```

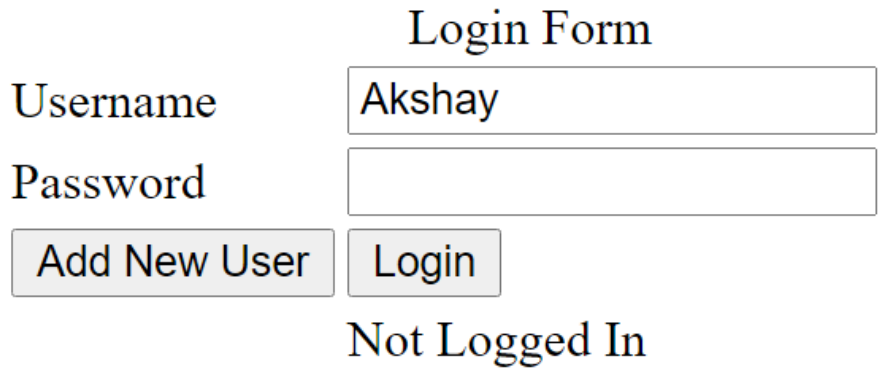
```
            int count = int.Parse(rdr[0].ToString());
            rdr.Close();
            if (count == 0)
            {
                cmd.ExecuteNonQuery();
                Label1.Text = "Record Inserted";
            }
            else
            {
                Label1.Text = "Record Already Exist";
            }
            con.Close();

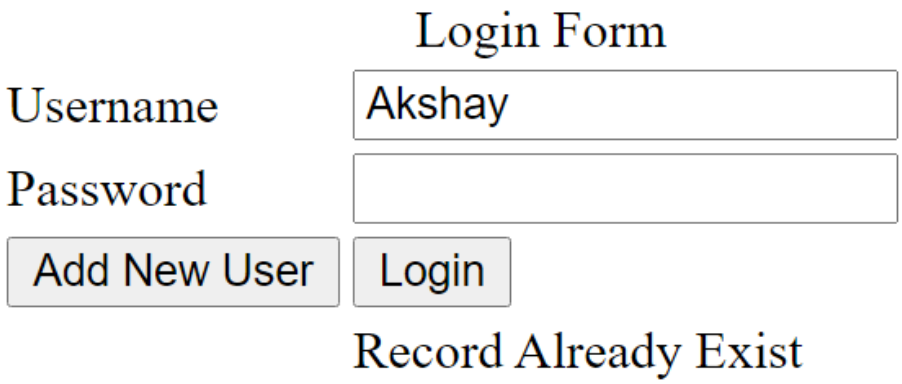
        }

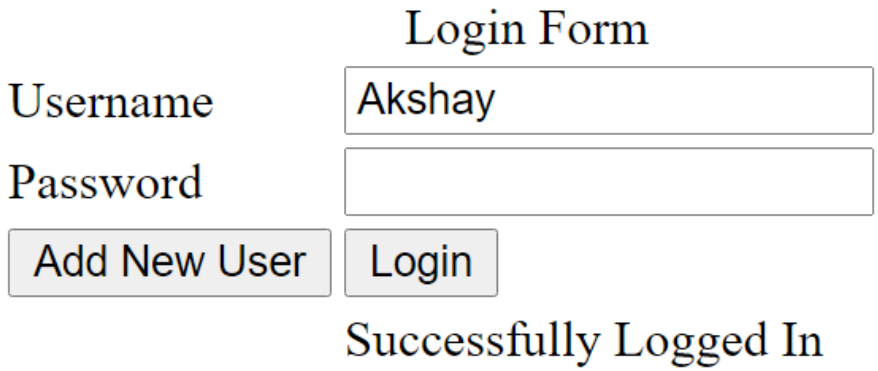
        protected void Button2_Click(object sender, EventArgs e)
        {
            SqlConnection con = new SqlConnection(Properties.Settings.Default.constr);
            string sql = "select count(*)from UserTable where Username=@uname and
            password=@passwd";
            SqlCommand cmd = new SqlCommand(sql, con);
            cmd.Parameters.AddWithValue("@uname", TextBox1.Text);
            cmd.Parameters.AddWithValue("@passwd", TextBox2.Text);
            con.Open();
            SqlDataReader rdr = cmd.ExecuteReader();
            rdr.Read();
            int count = int.Parse(rdr[0].ToString());
            rdr.Close();
            if (count == 0)
            {
                cmd.ExecuteNonQuery();
                Label1.Text ="Not Logged In";
            }
            else
            {
                Label1.Text = "Successfully Logged In";
            }
            con.Close();
        }
    }
}
```

**OUTPUT :**


Login Form
Username
Akshay
Password
Add New User
Login
Record Inserted


Login Form
Username
Akshay
Password
Add New User
Login
Not Logged In


Login Form
Username
Akshay
Password
Add New User
Login
Record Already Exist


Login Form
Username
Akshay
Password
Add New User
Login
Successfully Logged In

**Program 05: Create a web application to design an admission form with client side validations.**

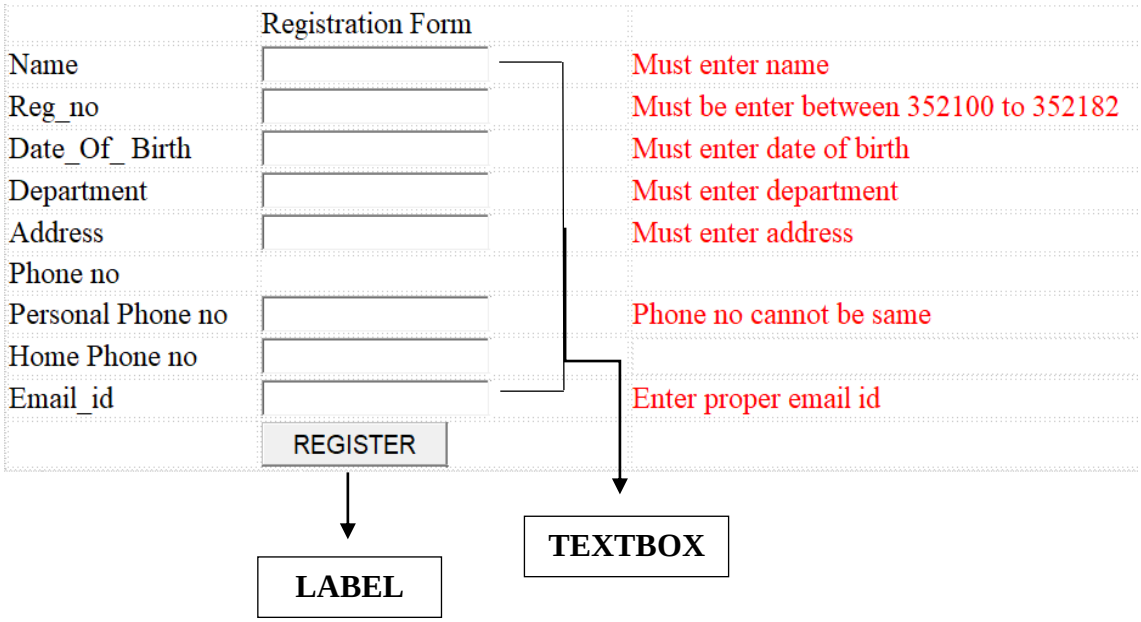
**Creating a Web Application:-**

Click on the icon of Visual Studio 2010 → New Project → Visual C# → Web → ASP.NET Web Application → Name the Web Application →Browse the location→ Ok.

**Creating a Web Form:**

Right click on the name of the Web Application → Add → New Item → Visual C# Web →Web Form → Add.

**DESIGN:**




| SL NO | CONTROLS | NAME | TEXT |
|---|---|---|---|
| 1 | TEXTBOX1 | TEXTBOX1 | ----- |
| 2 | TEXTBOX2 | TEXTBOX2 | ----- |
| 3 | TEXTBOX3 | TEXTBOX3 | ----- |
| 4 | TEXTBOX4 | TEXTBOX4 | ----- |
| 5 | TEXTBOX5 | TEXTBOX5 | ----- |
| 6 | TEXTBOX6 | TEXTBOX6 | ----- |
| 7 | TEXTBOX7 | TEXTBOX7 | ----- |
| 8 | TEXTBOX8 | TEXTBOX8 | ----- |
| 9 | BUTTON1 | BUTTON1 | REGISTER |

| SL NO | CONTROLS | ERROR MESSAGE | NAME |
|---|---|---|---|
| 1 | RequiredFieldValidator | Must enter name | RequiredFieldValidator1 |
| 2 | RangeValidator | Must be enter between 352100 to 352182 | RangeValidator1 |
| 3 | RequiredFieldValidator | Must enter date of birth | RequiredFieldValidator3 |
| 4 | RequiredFieldValidator | Must enter department | RequiredFieldValidator4 |
| 5 | RequiredFieldValidator | Must enter address | RequiredFieldValidator5 |
| 6 | CompareValidator | Phone no cannot be same | CompareValidator1 |
| 7 | RegularExpressionValidator | Enter proper email id | RegularExpressionValidator1 |

**SOURCE CODE:**

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="WebForm1.aspx.cs"
Inherits="c5555555555.WebForm1" %>

<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN"
"http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">

<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
    <title></title>
    <style type="text/css">
        .style1
        {
            width: 100%;
        }
        .style2
        {
            width: 207px;
        }
        .style3
        {
            width: 314px;
        }
    </style>
</head>
<body>
    <form id="form1" runat="server">
    <div>

        <table class="style1">
```

```
<tr>
   <td class="style2">
       </td>
   <td class="style3">
      Registration Form</td>
   <td>
       </td>
</tr>
<tr>
   <td class="style2">
      Name</td>
   <td class="style3">
      <asp:TextBox ID="TextBox1" runat="server"></asp:TextBox>
   </td>
   <td>
      <asp:RequiredFieldValidator ID="RequiredFieldValidator1" runat="server"
      ControlToValidate="TextBox9" ErrorMessage="Must enter name"
ForeColor="Red"></asp:RequiredFieldValidator>
   </td>
</tr>
<tr>
   <td class="style2">
      Reg_no</td>
   <td class="style3">
      <asp:TextBox ID="TextBox2" runat="server"></asp:TextBox>
   </td>
   <td>
      <asp:RangeValidator ID="RangeValidator1" runat="server"
         ControlToValidate="TextBox2"
         ErrorMessage="Must be enter between 352100 to 352182" ForeColor="Red"
         MaximumValue="352182"
         MinimumValue="352100"></asp:RangeValidator>
   </td>
</tr>
<tr>
   <td class="style2">
      Date_Of_Birth</td>
   <td class="style3">
      <asp:TextBox ID="TextBox3" runat="server"></asp:TextBox>
   </td>
   <td>
      <asp:RequiredFieldValidator ID="RequiredFieldValidator3" runat="server"
         ControlToValidate="TextBox3" ErrorMessage="Must enter date of birth"
         ForeColor="Red"></asp:RequiredFieldValidator>
   </td>
</tr>
```

```
<tr>
   <td class="style2">
      Department</td>
   <td class="style3">
      <asp:TextBox ID="TextBox4" runat="server"></asp:TextBox>
   </td>
   <td>
      <asp:RequiredFieldValidator ID="RequiredFieldValidator4" runat="server"
         ControlToValidate="TextBox4" ErrorMessage="Must enter department"
         ForeColor="Red"></asp:RequiredFieldValidator>
   </td>
</tr>
<tr>
   <td class="style2">
      Address</td>
   <td class="style3">
      <asp:TextBox ID="TextBox5" runat="server"></asp:TextBox>
   </td>
   <td>
      <asp:RequiredFieldValidator ID="RequiredFieldValidator5" runat="server"
          ControlToValidate="TextBox5" ErrorMessage="Must enter address"
         ForeColor="Red"></asp:RequiredFieldValidator>
   </td>
</tr>
<tr>
   <td class="style2">
      Phone no</td>
   <td class="style3">
       </td>
   <td>
       </td>
</tr>
<tr>
   <td class="style2">
      Personal Phone no</td>
   <td class="style3">
      <asp:TextBox ID="TextBox6" runat="server"></asp:TextBox>
   </td>
   <td>
      <asp:CompareValidator ID="CompareValidator1" runat="server"
         ControlToCompare="TextBox6" ControlToValidate="TextBox7"
          ErrorMessage="Phone no cannot be same" ForeColor="Red"
         Operator="NotEqual"></asp:CompareValidator>
   </td>
</tr>
<tr>
```

```
            <td class="style2">
                Home Phone no</td>
            <td class="style3">
                <asp:TextBox ID="TextBox7" runat="server"></asp:TextBox>
            </td>
            <td>
                 </td>
        </tr>
        <tr>
            <td class="style2">
                Email_id</td>
            <td class="style3">
                <asp:TextBox ID="TextBox8" runat="server"></asp:TextBox>
            </td>
            <td>
                <asp:RegularExpressionValidator ID="RegularExpressionValidator1"
                runat="server"
                ControlToValidate="TextBox8" ErrorMessage="Enter proper email id"
                ForeColor="Red"
                 ValidationExpression="\w+([-+.']\w+)*@\w+([-.]\w+)*\.\w+([-
                .]\w+)*"></asp:RegularExpressionValidator>
            </td>
        </tr>
        <tr>
            <td class="style2">
                 </td>
            <td class="style3">
                <asp:Button ID="Button1" runat="server" Text="REGISTER" />
            </td>
            <td>
                 </td>
        </tr>
    </table>

  </div>
  </form>
</body>
</html>
```

**OUTPUT:**

Registration Form

| | | |
|---|---|---|
| Name | | Must enter name |
| Reg_no | 152 | Must be enter between 352100 to 352182 |
| Date_Of_Birth | | Must enter date of birth |
| Department | | Must enter department |
| Address | | Must enter address |
| Phone number | | |
| Personal Phone no | 6363355885 | Phone no cannot be same |
| Home Phone no | 6363355885 | |
| Email_id | snadesh564gmail.com | Enter Proper email ID |

REGISTER

**Program 06: Design a webpage to enter Student information such as Student no, Student Name, marks in 3 subjects. Use the following buttons for**

- **Add -> for adding the record to the database (Insert at least 5 records). Calculate total, percentage, grade and store it.**
- **Display – Display the records from the database using data grid view.**

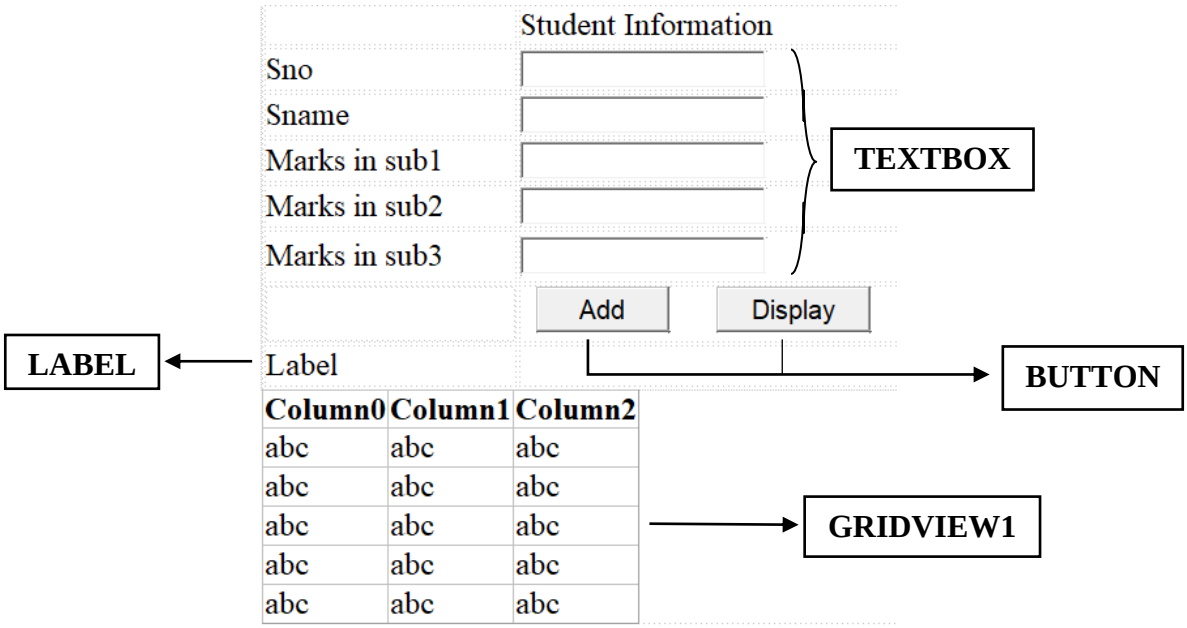
**Creating a Web Application:-**

Click on the icon of VisualStudio2010 → New Project → Visual C# → Web → ASP.NET Web Application→ Name the Web Application → Browse the location→ Ok.

**Creating a Web Form:**

Right click on the name of the Web Application → Add → New Item → Visual C# Web →Web Form → Add

**DESIGN:**




| SL NO | CONTROLS | NAME | TEXT |
|---|---|---|---|
| 1 | TextBox1 | TextBox1 | ----- |
| 2 | TextBox2 | TextBox2 | ----- |
| 3 | TextBox3 | TextBox3 | ----- |
| 4 | TextBox4 | TextBox4 | ----- |
| 5 | Button1 | Button1 | Add |
| 6 | Button2 | Button2 | Display |
| 7 | Label1 | Label1 | ----- |
| 8 | GridView | GridView1 | ----- |

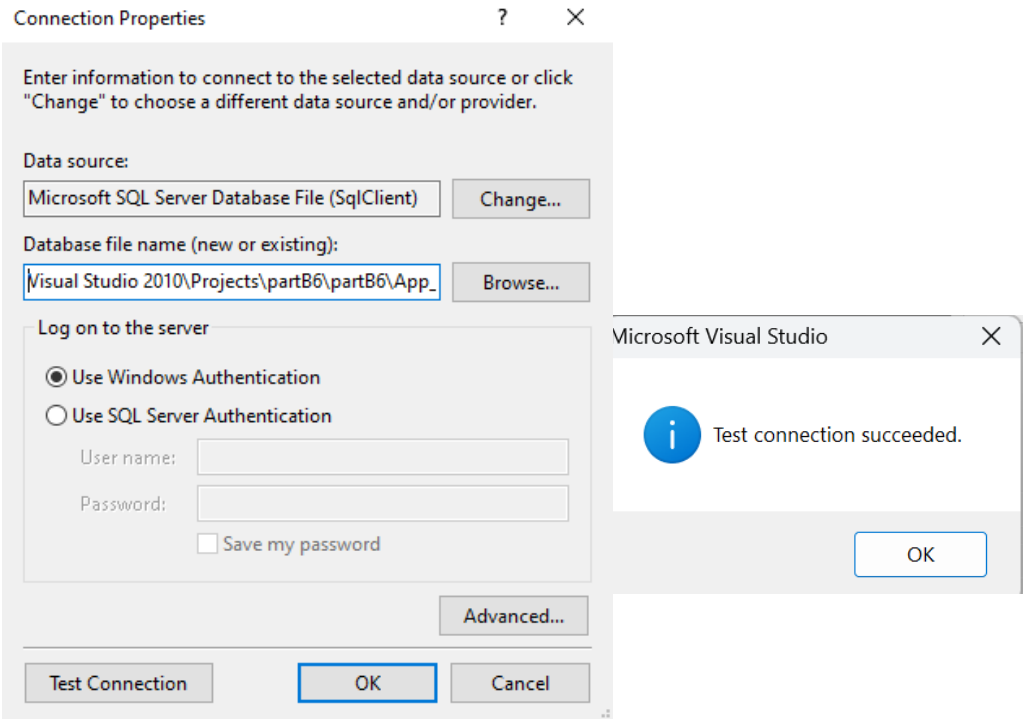
**Database:**

dbo.StdInfo: Tabl...TA\DATABASE1.MDF)*

| Column Name | Data Type | Allow Nulls |
|---|---|---|
| StudentNo | int | ☐ |
| StudentName | varchar(50) | ☑ |
| Sub1Marks | int | ☑ |
| Sub2Marks | int | ☑ |
| Sub3Marks | int | ☑ |
| TotalMarks | int | ☑ |
| Percentage | decimal(6, 2) | ☑ |
| Grade | varchar(50) | ☑ |

**Connection:**

Under Solution Explorer → Properties → Settings → Browse Microsoft SQL Server Database File (SqlClient) →Test Connection → Ok.




**SOURCE CODE:**

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;
using System.Data.SqlClient;
using System.Data;
```

```
namespace Stdunet_info
{
   public partial class WebForm1 : System.Web.UI.Page
   {
      protected void Button1_Click(object sender, EventArgs e)
      {
          try
         {
            int Sno = int.Parse(TextBox1.Text);
            string Sname = TextBox2.Text;
            int m1 = int.Parse(TextBox3.Text);
            int m2 = int.Parse(TextBox4.Text);
            int m3 = int.Parse(TextBox5.Text);
            int total = m1 + m2 + m3;
            double per = total / 3.0;
            string grade = "";
            if (per < 35)
               grade = "fail";
            else if (per < 60)
               grade = "second class";
            else if(per<75)
               grade = "first class";
            else
               grade = "distiction";
            SqlConnection con = new SqlConnection(Properties.Settings.Default.constr);
           string query="Insert into StdInfo values(@Sno,@Sname,
         @m1,@m2,@m3,@total,@per,@grade)";
            SqlCommand cmd = new SqlCommand(query, con);
            cmd.Parameters.AddWithValue("@Sno",Sno);
            cmd.Parameters.AddWithValue("@Sname", Sname);
            cmd.Parameters.AddWithValue("@m1", m1);
            cmd.Parameters.AddWithValue("@m2", m2);
            cmd.Parameters.AddWithValue("@m3", m3);
            cmd.Parameters.AddWithValue("@total",total);
            cmd.Parameters.AddWithValue("@per",per);
            cmd.Parameters.AddWithValue("@grade",grade);
            con.Open();
            cmd.ExecuteNonQuery();
            Label1.Text = "Student Record Inserted";
            con.Close();
            TextBox1.Text = "";
            TextBox2.Text = "";
            TextBox3.Text = "";
            TextBox4.Text = "";
            TextBox5.Text = "";
```

```
        }
        catch (Exception ex)
        {
            Label1.Text = "Error:" + ex.Message;
        }

    }

    protected void Button2_Click(object sender, EventArgs e)
    {
        try
        {
            SqlConnection con = new SqlConnection(Properties.Settings.Default.constr);
            string query = "select * from StdInfo";
            SqlDataAdapter da = new SqlDataAdapter(query, con);
            DataTable dt = new DataTable();
            da.Fill(dt);
            GridView1.DataSource = dt;
            GridView1.DataBind();
            Label1.Text = "Student details are listed below";
        }
        catch (Exception ex)
        {
            Label1.Text = "Error:" + ex.Message;
        }
    }
  }
}
```

**OUTPUT:**

Student Information

| | |
|---|---|
| Sno | 101 |
| Sname | Umesh |
| marks in Sub1 | 90 |
| marks in Sub2 | 95 |
| marks in Sub3 | 91 |

Add Dislpay

Student Record Inserted

Student Information

| | |
|---|---|
| Sno | |
| Sname | |
| marks in Sub1 | |
| mark in Sub2 | |
| mark in Sub3 | |

Add Display

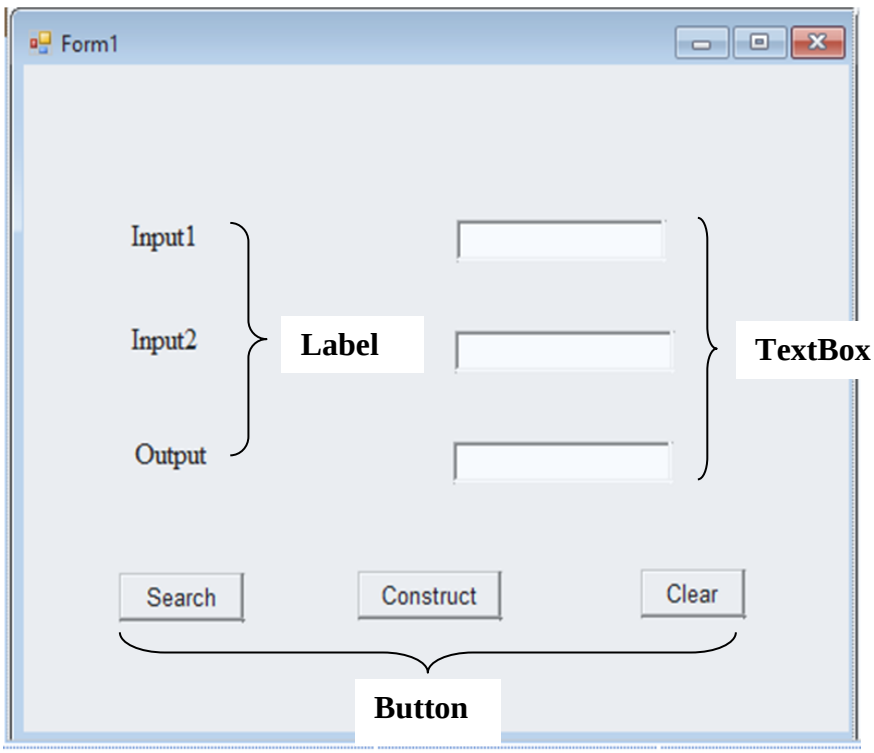
Students details are listed below

| sno | sname | sub1mark | sub2mark | sub3mark | total | per | grade |
|---|---|---|---|---|---|---|---|
| 101 | Umesh | 90 | 95 | 91 | 276 | 92.00 | distinction |
| 102 | Bhoja | 71 | 65 | 70 | 206 | 68.67 | first class |
| 103 | Shiva | 58 | 50 | 55 | 163 | 54.33 | second class |
| 104 | Akshay | 20 | 25 | 30 | 75 | 25.00 | fail |

**Program 07: Create ASP.NET web application with the given user interface to input two strings(str1, str2) and perform two operations "Search" and Construct" by clicking respective buttons. The result of both operations shall be displayed in output TextBox.**

**SERACH: If the user clicks on "Search" button then appearance of str2 is searched in str1 and removed from str1. Also, the characters of str1, before and after str2 are concatenated together. Ignore the cases where there is no character in str1 before or after the str2.**

**DESIGN:**




| SL NO | CONTROLS | NAME | TEXT |
|---|---|---|---|
| 1 | Label1 | Label1 | Input1 |
| 2 | Label2 | Label2 | Input2 |
| 3 | Label3 | Label3 | Output |
| 4 | TextBox1 | TextBox1 | ----- |
| 5 | TextBox2 | TextBox2 | ----- |
| 6 | TextBox3 | TextBox3 | ----- |
| 7 | Button1 | Button1 | Search |
| 8 | Button2 | Button2 | Construct |
| 9 | Button3 | Button3 | Clear |

**SOURCE CODE:**

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;

namespace partb7
{
   public partial class WebForm1 : System.Web.UI.Page
   {
      protected void Button1_Click(object sender, EventArgs e)
         {
            string str1 = TextBox1.Text;
            string str2 = TextBox2.Text;
            string result = "";

            if (str1.Contains(str2))
            {
               result = str1.Replace(str2, "");

               TextBox3.Text = result;
            }
            else
            {
               TextBox3.Text = str1;
            }
         }


      protected void Button2_Click(object sender, EventArgs e)
      {
         string str1 = TextBox1.Text;
         string str2 = TextBox2.Text;
         string result = "";

         int len1 = str1.Length;
         int len2 = str2.Length;
         int minLength = Math.Min(len1, len2);

         for (int i = 0; i < minLength; i++)
         {
            result += str1[i];
            result += str2[i];
```

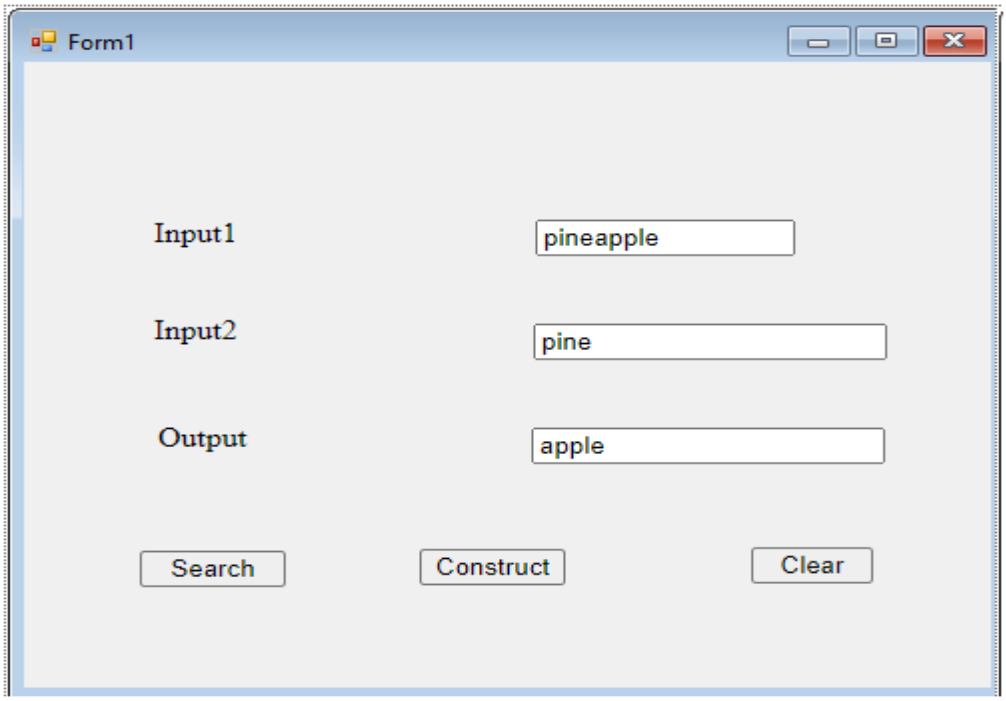
```
        }

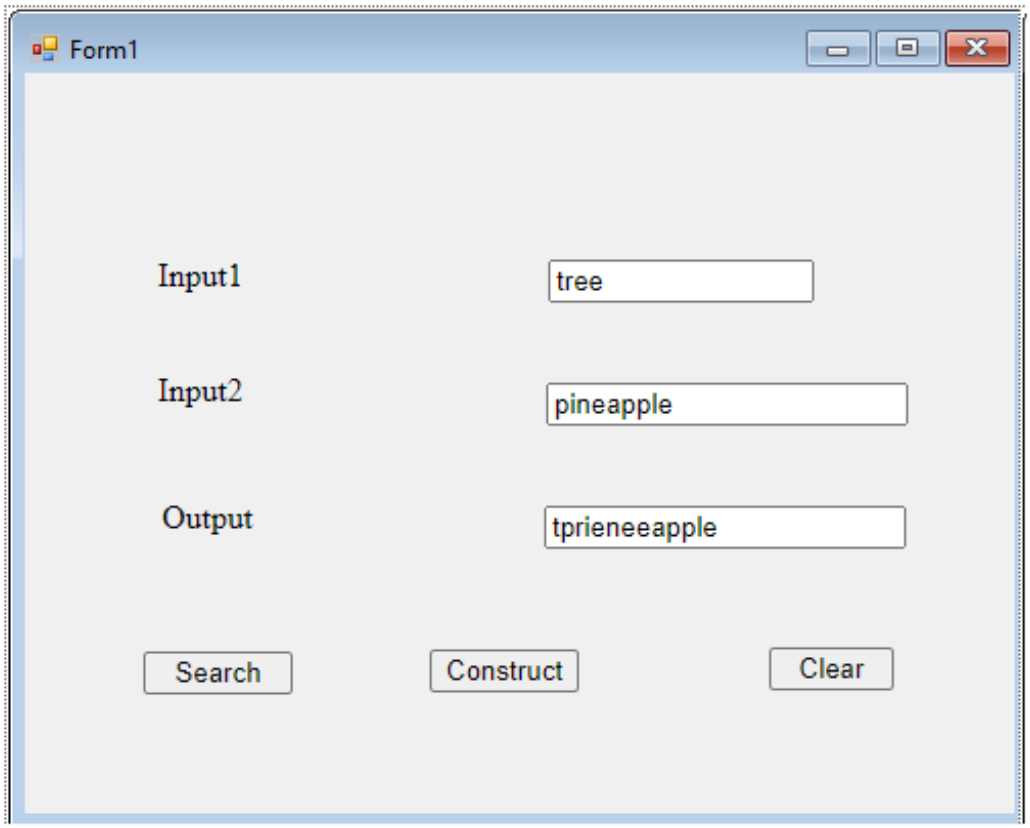
        if(len1>len2)
            result+=str1.Substring(minLength);
        else if(len2>len1)
            result+=str2.Substring(minLength);

        TextBox3.Text=result;
    }

    protected void Button3_Click(object sender, EventArgs e)
    {
        TextBox1.Text = "";
        TextBox2.Text = "";
        TextBox3.Text = "";
    }
  }
}
```

**OUTPUT:**




Form1
Input1
tree
Input2
pineapple
Output
tprieneeapple
Search
Construct
Clear